

# Support Vector Machines (SVM) - Primal and Dual Formulations

## Introduction

Support Vector Machines (SVM) are a popular technique for supervised learning, used primarily for classification tasks. The goal of SVM is to find the optimal hyperplane that separates the data points of different classes with the maximum margin. We will discuss the primal and dual formulations of SVM and explain how to solve the dual form using quadratic programming.

## Primal Problem Formulation

Given a training set of  $N$  examples:  $\{(x_i, y_i)\}_{i=1}^N$ , where  $x_i \in \mathbb{R}^d$  and  $y_i \in \{-1, 1\}$ , the primal form of the SVM optimization problem with a soft margin is:

### Objective:

$$\min_{w, b, \zeta} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \zeta_i$$

### Subject to:

$$\begin{aligned} y_i(w \cdot x_i - b) &\geq 1 - \zeta_i \\ \zeta_i &\geq 0 \quad \forall i \end{aligned}$$

where:

- $w$  is the weight vector.
- $b$  is the bias term.
- $\zeta_i$  are the slack variables for each data point.
- $C$  is the regularization parameter that controls the trade-off between maximizing the margin and minimizing the classification error.

## Dual Problem Formulation

The dual form of the SVM optimization problem can be derived using the Lagrangian method. Here's the step-by-step process to convert the primal problem to the dual problem:

### 1. Construct the Lagrangian:

The Lagrangian  $L(w, b, \zeta, \alpha, \mu)$  for the primal problem is:

$$L(w, b, \zeta, \alpha, \mu) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \zeta_i - \sum_{i=1}^N \alpha_i [y_i(w \cdot x_i - b) - 1 + \zeta_i] - \sum_{i=1}^N \mu_i \zeta_i$$

Here,  $\alpha_i \geq 0$  and  $\mu_i \geq 0$  are the Lagrange multipliers.

## 2. Take Partial Derivatives and Set Them to Zero:

To find the saddle points, take the partial derivatives of  $L$  with respect to primal variables  $w$ ,  $b$ , and  $\zeta$  and set them to zero.

- Partial derivative with respect to  $w$ :

$$\frac{\partial L}{\partial w} = w - \sum_{i=1}^N \alpha_i y_i x_i = 0 \quad \Rightarrow \quad w = \sum_{i=1}^N \alpha_i y_i x_i$$

- Partial derivative with respect to  $b$ :

$$\frac{\partial L}{\partial b} = - \sum_{i=1}^N \alpha_i y_i = 0 \quad \Rightarrow \quad \sum_{i=1}^N \alpha_i y_i = 0$$

- Partial derivative with respect to  $\zeta_i$ :

$$\frac{\partial L}{\partial \zeta_i} = C - \alpha_i - \mu_i = 0 \quad \Rightarrow \quad \alpha_i + \mu_i = C$$

Since  $\mu_i \geq 0$ , this implies:

$$0 \leq \alpha_i \leq C$$

## 3. Substitute Back into the Lagrangian:

Substitute  $w = \sum_{i=1}^N \alpha_i y_i x_i$  back into  $L$ :

$$L(\alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

## 4. Formulate the Dual Problem:

The dual form of the SVM optimization problem is:

**Objective:**

$$\max_{\alpha} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

**Subject to:**

$$0 \leq \alpha_i \leq C$$

$$\sum_{i=1}^N \alpha_i y_i = 0$$

## Solving the Dual Problem Using Quadratic Programming

To solve the dual problem, we can use Quadratic Programming (QP). Here's how to set up the QP problem:

### Formulating the QP Problem:

The QP problem in standard form is:

$$\min_{\alpha} \frac{1}{2} \alpha^T P \alpha - q^T \alpha$$

subject to:

$$G\alpha \leq h$$

$$A\alpha = b$$

Where:

- $P = (y_i y_j x_i \cdot x_j)$ .
- $q = -1_N$  (vector of -1s).
- $G = \begin{bmatrix} -I \\ I \end{bmatrix}$  (stacked identity matrices).
- $h = \begin{bmatrix} 0 \\ C \end{bmatrix}$  (stacked 0s and Cs).
- $A = y^T$  (transpose of label vector).
- $b = 0$ .

### Using a QP Solver:

Once the QP problem is formulated, various QP solvers can be used to find the solution. Libraries like `cvxopt` in Python are commonly used for solving such optimization problems.

### Finalizing the Classifier:

After solving for  $\alpha$ , the weight vector  $w$  and bias term  $b$  can be computed as follows:

**Weight Vector:**

$$w = \sum_{i=1}^N \alpha_i y_i x_i$$

**Bias Term:**

Choose a support vector  $(x_k, y_k)$  such that  $0 < \alpha_k < C$ :

$$b = y_k - w \cdot x_k$$

## Decision Function for a New Data Point:

The decision function for a new data point  $x$  is:

$$f(x) = w \cdot x - b$$

The class prediction is:

$$\text{class} = \text{sign}(f(x))$$

## Conclusion

This document provides an overview of solving SVMs for linearly separable data with soft margin penalties. We discussed the primal problem, converted it to the dual form, and detailed how to solve the dual problem using quadratic programming. The dual formulation is often computationally more efficient, especially when incorporating kernel functions for non-linearly separable data.